



JABchem



Not to be shared without the copyright holder's permission

Nat5 Chemistry

Nat5

Question Bank

**Self-Evaluation
Question Bank**

Nat5 Traffic Lights			Past Paper Question Bank Unit 1.1 Reaction Rates														Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027			
1	L1a L1c	L1c			L3b(ii)		L5d		L2a L2b(ii)	L2a L2c L2d		L1a(iii)A						
2a 2b 2c					mc2 L3b(iii)		mc1		mc2 L3b(iii)				L1b(ii) L1b(iii)					
2d 3	L13c									L9c(iii)A	L4b	L1a(ii)	L4b(ii)					
4	L1b	L1b	mc1	L1a	L3b(i)	mc1	L1b(i)	mc1 mc2	L3b(i)	L2b	mc1	mc1 L1a(iii)B L1a(iii)C	L1b(i)					
5																		

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.1 Reaction Rates															Copyright JAB chem
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015	
1				L7b(i)								L4b(ii)	L2a(ii)		L1b(i)		
2a 2b 2c		mc5 L14c		mc2	mc4	mc1 L3d		mc2	mc1 L15c	mc3	mc1	mc3	mc2 L4b			mc6	
2d 3							L7b(iv)				L5c(ii)						
4	mc5	mc7	L13b(ii)	L7b(ii)	mc3	L3c		L2a			mc2	L4b(iii)	L2a(i)	L3b	L1b(iii)	mc5	
5																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.1 Reaction Rates														Copyright JAB chem	
Outcome	2000 <u>Credit</u>	2001 <u>Credit</u>	2002 <u>Credit</u>	2003 <u>Credit</u>	2004 <u>Credit</u>	2005 <u>Credit</u>	2006 <u>Credit</u>	2007 <u>Credit</u>	2008 <u>Credit</u>	2009 <u>Credit</u>	2010 <u>Credit</u>	2011 <u>Credit</u>	2012 <u>Credit</u>	2013 <u>Credit</u>			
1					12a					13b(i)			11a				
2a 2b 2c		2a			4a,4b 12c		12c	2a 2b	2a	13c	12d		11b(i) 11b(ii)	18a(ii)			
2d 3	11c(ii)					12b(ii)						16b(i)					
4													13b(ii)				
5																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.1 Reaction Rates														Copyright JAB chem	
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General			
1		16a 16b		18b(i)													
2a 2b 2c	7a	16c	7a 7b	8		2a 2b	2 9b	13c	13a(ii)	3a 3b		2a 2b 19a(i) 19b(ii)	2a 2b	2a 2b			
2d 3		11a 16d			17a(i) 17a(ii) 17b	9c		14c(i)	16c 18b(i)	13a	15c(iii)			15c			
4																	
5																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.2a Periodic Table and Atoms														Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027		
6 7								mc3		mc1	L3a	L5	mc1				
8									mc1		L1c		L2b				
9					mc3												
10																	
11	mc2	mc2			L1a(i) L1a(ii)		mc2				mc2		L8d				
12										mc2							
13																	
14	L2a	L2a	L1b(ii)	mc2		L1a			L1a(i)	L9a		mc2					
15a	L2b	L2b	L1b(i)	mc1	L5b	mc2 L1b	L7b		mc2 L1b			mc3	mc2 L8b				
15b	mc5	mc5		mc3				mc4			L1d						
16	L2c	L2c			L1b	L1c	L6a(i) L6a(ii)		L1a(ii)		L1b						

Nat5

Traffic Lights

Past Paper Question Bank

Unit 1.2a Periodic Table and Atoms

Copyright

JAB

chem

Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015
6 7			mc1						mc3							mc1
8	L3a							mc1						L1a(ii)		mc2
9		mc2				mc4				L1b(i)			mc4			
10										L1b(ii)	L1a					
11								L1a				mc4		mc5 L4a		
12				mc3		mc5		L1b(i)	mc4		L1b(ii)					mc7
13							mc3				mc9 L1b(i)					
14	mc2	L1b		mc5			mc5	mc4	L1b(ii)	mc4			L1c(ii)		L2b(ii)	L1b
15a	mc1	L1a	mc3	mc24	mc5 L1a(i) L1a(ii)	L1b			L1b(i)	L1a		mc5	L1c(i)		L2b(i)	L1a
15b			mc6				mc7 L3c	L13a(i)	mc6		L9d		mc5	L11a		
16	L3b				L1b									L4b		

Nat5 Traffic Lights Past Paper Question Bank Unit 1.2a Periodic Table and Atoms Copyright JAB chem																
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit		
6 7																
8							5a									
9	1a	3c		18a	6b	1b	5b			2b	3b	3b		2a		
10																
11		6			7			7	15a	7		7	9	7		
12				1a												
13				1b												
14	10b	3b	5b	11a(i)	6a		9a(i)			9b		11b(i)	4c			
15a	10a		6				9a(iii)		15b(i)	9a	3a	11b(ii)		10a(i) 10a(ii)		
15b		3a	5a	11c		13b		10b					4a			
16	10c			11a(ii)		13a(i) 13a(ii)	9a(ii)	10a		9c				10b		

Nat5 Traffic Lights Past Paper Question Bank Unit 1.2a Periodic Table and Atoms Copyright JAB chem																
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General		
6 7																
8	1a			3b		10a(ii)		1b		1c		1b	1c			
9		8a		3a			10b(i)		2b	1a						
10							10a						10a			
11																
12		8b								10a		9a				
13																
14																
15a																
15b																
16																

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.2b Covalent Bonding Copyright JAB chem													
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027
17 18					mc5					mc6			L2d(ii)		
19 22						L3a		L3a					L2a		
20		mc1					mc3			L9c(i)	L1a	L7c	L3a(ii)		
21	mc4 L7b(i)	mc4 L7b(i)	L7a(ii)	L4a	L1c(i)	L3b	mc4 L10b	mc6	mc3 L4b	mc5 L4a(ii)	mc3 L7d	L2a(i) L2a(ii)	L2d(i)		
23															
24	L7b(ii)	L7b(ii)		mc5 L14a(ii)	L11c		L4c		L1c	mc3 L9c(ii)	L7e L11a	mc4 L3e	L5d(ii)		
25															
26	mc1						mc6								
27															

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.2b Covalent Bonding															Copyright JAB chem
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015	
17 18							L2a										
19 22			L4b													L3b	
20			mc2								mc5						
21	L4c				L4a		L2b		L5a		L3b(i) L3b(ii)	L2a L2b	mc9	mc6	L4a	L2a	
23																	
24					L13a			L13b(ii)							L4b	L2b(i) L5a	
25																	
26			L2b(ii)					mc6			L3a		L1a	L6b			
27			L2b(i)														

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.2b Covalent Bonding														Copyright JAB chem	
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit			
17 18	1b	16c		11b	13a		4b	15b	10b	12a	15c	5c		17b			
19 22						15a							13a				
20																	
21	13d		3b	10c	13b(i) 13b(ii)		9b	15a		12b	2b	3c		17a			
23																	
24			2b 12a(ii)			4b		5a			14a	4b					
25																	
26										3a							
27																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.2b Covalent Bonding														Copyright JAB chem	
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General			
17 18		18a(ii)	13b			1b		1c	1d		10c	10b	6b				
19 22																	
20	6b	18a(i)	2b	6c	5b	9a	6b		2a 8	14b	15a	5b	1a	3a			
21	6a																
23																	
24											5a						
25																	
26																	
27																	

Nat5 Traffic Lights			Past Paper Question Bank Unit 1.2c Ionic Bonding														Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027			
28										L9b								
29 30				L11a			mc5	mc7		L12c(i) L12c(ii)								
31													mc4					
32							L7a				mc4							
33					mc5													
34									mc20									
35	mc3	mc3		L9b(ii)			L7b(i)		L9b(iii)				mc5					
36												mc25						
-			L2a		mc6	L3d	mc6	mc8			mc17	mc18						

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.2c Ionic Bonding															Copyright JAB chem
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015	
28				L12b													
29 30				mc6													
31																	
32										mc9							
33																	
34																	
35													mc10				
36																	
-						mc7	mc6		mc10			L1a	mc7		mc7		

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.2c Ionic Bonding														Copyright JAB chem	
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit			
28																	
29																	
30																	
31																	
32																	
33																	
34													18a				
35		18d							13c	10c							
36								19c(i)									
-		7	2a	9a+b 13a+b	5a 5b			5b		3b				8			

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.2c Ionic Bonding														Copyright JAB chem	
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General			
28																	
29																	
30																	
31																	
32													18a				
33																	
34																	
35	18b					15b		17b			19b		18b				
36																	
-			8				8		9		7a 7b	8		8			

Nat5 Traffic Lights			Past Paper Question Bank Unit 1.3a Chemical Formulae														Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027			
37		L6e						L8b				L7b	L7b(i)					
38			L11c					L8c					mc3					
39					mc7		L15c		L10b	L9c(i)	L7a							
40							L5c		mc6									
41																		
42																		
43											L3b(i)		mc9					
44 45	L5a	mc23 L3a	mc3 L3d	mc6 L6b		mc3		L11a	L7c(ii)	L11a(ii)	mc5	L4c(iii)						

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.3a Chemical Formulae															Copyright JAB chem	
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015		
37						mc2					L11b(i)	mc2						
38			L1a															
39	L5b			L13c				L11c		mc11 L14a		L4a(ii)	L1b	mc7				
40																		
41																		
42																		
43		L4a	mc4	mc8	L14d	L2a(i)	mc8		L15a		mc7			L6c	L14b			
44 45		mc6	mc5			L3a						mc8			mc6	mc9 L4b(ii)		

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.3a Chemical Formulae														Copyright JAB chem	
Outcome	2000 <a>Credit	2001 <a>Credit	2002 <a>Credit	2003 <a>Credit	2004 <a>Credit	2005 <a>Credit	2006 <a>Credit	2007 <a>Credit	2008 <a>Credit	2009 <a>Credit	2010 <a>Credit	2011 <a>Credit	2012 <a>Credit	2013 <a>Credit			
37		13a(i)						17a						11a(i)			
38	1c 13a		3a		3c	1c	5c			4c	2c	16b(ii)	7c				
39							11c										
40								17d									
41																	
42																	
43																	
44 45		18b	14d		17a			19b	20b	10b	13c 16a(i)	18c	15d	16b(ii)			

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.3a Chemical Formulae														Copyright JAB chem	
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General			
37	5a		5a	2	14a	10c	13a	4c	10c	6a	13a		6a	4a			
38				12b				14c(ii)		10b(i)			16c(ii)	19a			
39																	
40							16b		15b(i)								
41																	
42																	
43						16b											
44																	
45																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.3b Mole Calculations & Balanced Equations												Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027
46	L4a(i)	L5a(i)	mc4	mc8 L14a(i)	mc1 L5c(i) L6b	mc7	L10a(i)		mc4 L8a(iii)	L6b(ii)	L11b(i)	L1a(i)	L9b		
47							L13a(ii)	L2d	L4c			mc8			
48a			L3b	mc9		mc8					L4c				
48b	L13a	L15a		L7b		L2c							L8c L11b		
48c															
49								mc9		mc4			L10a(i)		
50	L13b	L15b		L7b	L12d	mc9	mc7 L14b(ii)		mc5		mc6	L9a(iv)	L11b		
51	L4b(i)	L5b(i)	L10b		L3c(ii)	L12c	L16c	L5c(i)	L12c	L11c	L7f	mc7			
52	L11a	L12a	L12b		L4b	L10b		mc10	L3c(i)	L3e(ii)	L8d	L9d	mc8		

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.3b Mole Calculations & Balanced Equations															Copyright JAB chem	
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015		
46	mc25 L11b	L6c(ii)	L4a	L2b	mc2 L2a	mc8 L5a	L1b(i)	L4a(i) L8a	L3a	mc5 L4a	mc8 L2b	L3a	L2b	mc2 mc8	mc9	Mc11 L5b(i)		
47	L5c		mc7	mc29			mc10							mc9	mc10	mc10		
48a																		
48b		mc8			L5b	L4c(ii)	mc9	L14b	mc9				L4c	L6a				
48c																		
49							mc2	mc5	mc2	mc2	L2c			mc3	mc2	mc4		
50		L4b(ii)	mc22	mc17	mc21	mc21	L13a(i)	mc21	mc23	L12c(i)			L15a(i)	mc21	L10a			
51	L6b	L9b	L12b L13a	L9d	L8b	L13c	L5b L13a(ii)	L4b	L11a	L4b	mc20 L4b	L3c L5b	L15a(ii)	L12c	L4c(ii)	L12b		
52																		

Nat5 Traffic Lights Past Paper Question Bank Unit 1.3b Mole Calculations & Balanced Equations Copyright JAB chem																
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit		
46	13e	12a	12a(i)	10b(i)	17b(i)	16a	16a	10c(i)	12b	16b	14b(i)	20a(i)	15a			
47																
48a																
48b				19b												
48c					11b(ii)					16d			21c(ii)			
49									6			1				
50		14c	19a(ii)	19b	21b(ii)			20a	14c			20c(ii)	17b	18b(i)		
51	18b	12c	15b(ii)	12c		20b	15a		20a	19c	12e	17b		17c		
52		13c(ii)	12b	15c	11b(i) 17c	11b	11b	17c	16a	14b	16a(ii)		15c 21c(i)	16b(i)		

Nat5 Traffic Lights Past Paper Question Bank Unit 1.3b Mole Calculations & Balanced Equations Copyright JAB chem																
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General		
46																
47																
48a																
48b																
48c																
49					14b						18a			10c(i)		
50																
51																
52																

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.4a pH Copyright JAB chem													
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027
53		L6b					L13b		L7a(i)			L8b(iv)			
55 56							L10a(ii)								
54 57 58	mc7 L3a	mc7 L4a	mc5	L4b	L5d		mc8		mc9	L3b	L11b(ii)				
59 60							mc9	mc11	mc8			mc9	mc7		
61				mc10	mc8	L11a									
62			mc6	mc10							mc7	mc6	mc6		
63				mc11		mc5		mc12							

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.4a pH															Copyright JAB chem	
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015		
53								L9b			L10a			L13a(i)				
55 56	mc19					L13b(ii)				mc24	mc17				mc20			
54 57 58		mc19	mc20			mc27 L13b(i)	mc23	mc20	mc21		mc19			mc18		L15a		
59 60			mc17			mc19		mc19				mc22		mc20				
61	mc18				mc23			mc18		mc21			10c	mc23		mc24		
62					mc23			mc18	mc22	mc21		mc21	mc21		mc22	mc24		
63			mc21		mc24		mc24				mc22			mc22		mc25		

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.4a pH														Copyright JAB chem	
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit			
53																	
55 56																	
54 57 58												14a		2c			
59 60														5			
61							13c(ii)		4b								
62							4a		4c			5b					
63																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.4a pH														Copyright JAB chem	
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General			
53	7b 16a 16b		13c		12a	5a 5b	18a	11c	11v 21a		3a	17b(i) 17b(ii)	13a(i) 13a(ii)				
55 56																	
54 57 58				11b	4c			11e					13b	16a			
59 60							4	11d				17c					
61		9a(i)											17c	4c			
62		9a(ii)									4b			4b			
63																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.4b Neutralisation Reactions														Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027		
64										L8e	mc8	L9a(ii) L9a(iii)					
65a				L4c						L8a							
65b			L7c(i)							L8a							
65c	L4a(ii)	L5a(ii)					L1a		mc7	L5 L8a			L11c				
66							L10d	L2c L11b					L11a				
67		mc24	mc8	mc18		mc6		mc25	mc18		mc9	L8a(i)					
68																	
69																	
70			L13b	L15b					L13	L7a(i)	mc25	L8b(iii)	L10b(iii)				

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.4b Neutralisation Reactions															Copyright JAB chem
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015	
64											mc23						
65a		mc20			L8a			L3b(i)									
65b			L7b			L14b											
65c	L11c	mc24		mc28		L2a(ii)		mc22	L12a	mc22 L13c		mc24	L11b		mc23	mc26	
66								L15a	L13b						L11a	L14a(i)	
67		L12a(i)	L9a	mc21	L6a	mc24	L11b(i) L11b(ii)	mc25		L15a	mc26	mc25	L13b(i) L13b(ii)	L2d			
68													mc22				
69																	
70	L16	L12b	L8d	L8c	L16b	mc22				L12c(ii)	L13b	L11c(iii)				L15b(iii)	

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.4b Neutralisation Reactions														Copyright JAB chem	
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit			
64			1b		3b	20a				4a			7a				
65a																	
65b	5a																
65c	5a								5a	13a				18c			
66		13b			12d			17b									
67			13a	15a					17b(i)		17a		3	14a			
68								6b				6a					
69					21a			20b(i) 20b(ii)				20c(i)					
70			19b		21b(iii)		18b				19b	20d		18b(ii)			

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.4b Neutralisation Reactions														Copyright JAB chem	
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General			
64																	
65a			16a	16b													
65b										6b	8		20a(ii)				
65c	17a 17b	5b	5b			3b											
66																	
67																	
68																	
69																	
70																	

Nat5		Past Paper Question Bank												Copyright JABchem	
Traffic Lights		Unit 1.4c Preparation of Salts by Neutralisation													
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027
74															
75							mc25	L11c		L8c		L9a(i) L9b L9c			
76															

Nat5 Traffic Lights		Past Paper Question Bank Unit 1.4c Preparation of Salts by Neutralisation														Copyright JABchem	
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015	
74																	
75	L15a			L9a						L13a							
76	L15b			L9b L9c						L13b							

Nat5																Past Paper Question Bank																Copyright JAB chem	
Traffic Lights		Unit 1.4c Preparation of Salts by Neutralisation																															
Outcome	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013																			
	Credit	Credit	Credit	Credit	Credit	Credit	Credit	Credit	Credit	Credit	Credit	Credit	Credit	Credit																			
74							18c																										
75						14a																											
76						14b																											

Nat5																Copyright JAB chem	
Traffic Lights																	
Past Paper Question Bank																	
Unit 1.4c Preparation of Salts by Neutralisation																	
Outcome	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013			
	General	General	General	General	General	General	General	General	General	General	General	General	General	General			
74																	
75	17c		16b	7b	8a												
76																	8b

Nat5 Traffic Lights		Past Paper Question Bank														Copyright JAB chem	
Unit 2.1 Naming and Drawing Hydrocarbons																	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027		
1	L7a	mc15 L7a	L9a			L9a	L16a(i)	L7a	L8a(ii)	L10a		mc15 L4a	mc11 mc12				
2			L9b					L5c(ii)									
3							L4a		L12a		L2a L2c(i)	L10a					
4							L9a(i)				mc15						
5	mc13			L3b(ii)		mc11	L9a(ii)	L5a	L4f	mc12	L2b(ii)	L3d	L5b(ii)				
6a		L10a			L8a		mc10						mc14				
6b		mc13		L12b		L9b		mc13			mc11		L9a(i)				
7	mc12		L8b(ii)		mc12		mc11										

Nat5 Traffic Lights		Past Paper Question Bank														Copyright JAB chem	
Unit 2.1 Naming and Drawing Hydrocarbons																	
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015	
1													L5a				
2														L8b			
3																	
4						mc11											
5		L5b(i)		L14a			mc16	L6a				L11c(i)	L8a		L8a(ii)		
6a										mc16							
6b							mc13										
7	L1b	mc11	mc10	mc9	mc16	mc10		mc16	mc15					mc10	L7a(iii)	mc16	

Nat5 Traffic Lights Past Paper Question Bank Unit 2.1 Naming and Drawing Hydrocarbons Copyright JAB chem																
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit		
1	18a	17a		3c	20a		2b			19b	6b	22a				
2																
3																
4																
5		10a 10b	4a		2c	5b	4b		3b		6a	16a				
6a																
6b	3b	10c	4b	3b	2a 20d	5c	2a	16b(ii)	19b	5a	6c		5b	6c		
7																

Nat5 Traffic Lights Past Paper Question Bank Unit 2.1 Naming and Drawing Hydrocarbons Copyright JAB chem																
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General		
1																
2																
3		6b	10a			11a		9a					7a			
4			3b	10c(i)												
5			6b	5c			14b(ii)	14b(ii)		13b(I i)		20c	20b	15b		
6a																
6b																
7																

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.1b Alkanes														Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027		
8					mc10	mc10		L7b(i) L7b(ii)			L7b						
9a 10a												L10d(i)A					
9b 10b	mc8	mc8	mc10	mc12			L4b	mc14	mc13	L6a			L5b(i)				

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.1b Alkanes														Copyright JAB chem	
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015	
8					mc12						mc12	mc12					
9a 10a																	
9b 10b				L11b	mc11		L8a(i)		mc12					L8c	L7a(ii)		

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.1b Alkanes														Copyright JAB chem	
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit			
8						12a											
9a 10a																	
9b 10b									19a								

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.1b Alkanes														Copyright JAB chem	
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General			
8		3a 3b		10b			3a		4a		17a		12b 12c				
9a 10a	9a				2b 16a			3a 14a				20b					
9b 10b																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.1c Alkenes														Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027		
11						L12a			mc11 mc14 L12b(ii)								
12a 13a																	
12b 13b				L12a					mc16		mc10	mc12 L10c(i)					
14a 15a					mc11			L5b(i)	L12b(i)		mc12 L2b(i)		L4b(i)				
14b 15b		L10c			L8c(i) L8c(ii)			L5b(ii)			mc12						
14c 15c	L8b(i) L8b(ii)	L8c(i) L8c(ii)		L12c			mc12	mc15		L4a(i) L4a(ii)	mc12	L10d(i)B	mc15				

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.1c Alkenes															Copyright JAB chem
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015	
11			L12a(i)						mc13							mc17	
12a 13a	mc14					mc12						mc13					
12b 13b																	
14a 15a																	
14b 15b			L12a(ii)								L5d						
14c 15c	mc10	mc13			mc14	L6a(ii)				mc17			L7a			L6b(i) L6b(ii)	

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.1c Alkenes														Copyright JAB chem	
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit			
11																	
12a 13a													5c				
12b 13b																	
14a 15a				3a			6a			5c		22c	5a	6b			
14b 15b				14a(i) 14a(ii)	20c								20c				
14c 15c			16c(i) 16c(ii)					16b(i)									

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.1c Alkenes														Copyright JAB chem	
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General			
11		3b				7a		3b									
12a 13a	2b					7b	14b(i)		4c								
12b 13b																	
14a 15a		5a				14a(i) 14a(ii)											
14b 15b							14a				6c						
14c 15c				15b													

Nat5 Traffic Lights			Past Paper Question Bank Unit 2.1d Cycloalkanes														Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027			
16	mc9	mc9								mc14		mc11	mc10					
17 18			mc9									L10c(ii)						

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.1d Cycloalkanes															Copyright JAB chem
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015	
16					mc8	mc14		L6c	mc14	mc15			L8c	mc12			
17 18	mc16																

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.1d Cycloalkanes														Copyright JAB chem	
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit			
16																	
17 18																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.1d Cycloalkanes														Copyright JAB chem	
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General			
16																	
17																	
18																	

Nat5 Traffic Lights			Past Paper Question Bank Unit 2.2a Alcohols														Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027			
19																		
20																		
21											mc14							
22			mc13				mc14											
23	L8a	L8a	L8a	L3a(i)	L12a	L12a		L2a	L4d	L10b(i) L10b(ii)		mc5 L4c(i)						
24																		
25a 26a						L14a(i)	mc13						L9a(ii)					
25b 26b				mc13						L10b(iii)								

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.2a Alcohols															Copyright JAB chem	
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015		
19																		
20																		
21																		
22			mc11															
23			L5b(ii)			mc11			L6a		L9c			L13a(ii)				
24																		
25a 26a						L6a(i)	L7a					L9c	L7c			L6a		
25b 26b																		

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.2a Alcohols														Copyright JAB chem	
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit			
19																	
20																	
21																	
22																	
23																	
24														15a			
25a 26a											20a(i)						
25b 26b																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.2a Alcohols														Copyright JAB chem	
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General			
19																	
20																	
21																	
22																	
23																	
24																	
25a 26a																	
25b 26b																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.2b Carboxylic Acids Copyright JAB chem													
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027
27						mc13				L4b(ii)					
28								mc17							
29						L7b(ii)			mc17						
30	L6c L8d(ii)	L6d L8d(ii)		L13a		L7a	L13a(i)			mc12 L4b(i)	L6a(i)	L4c(i)	L4a		
31			L8b												
32 33	L8d(i)	L8d(i)		L3b(i)		L7b(i)			L4e(i)	mc11		mc13 mc14 L4c(ii)	mc13		
34									L7a(i)	mc8	mc16				

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.2b Carboxylic Acids															Copyright JAB chem
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015	
27																	
28																	
29																	
30				L1b			L8a								mc13	L8a	
31																	
32 33	L14a		L5b(iv)			L10a				L10c			L9b(i)				
34													mc14				

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.2b Carboxylic Acids														Copyright JAB chem	
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit			
27																	
28																	
29																	
30																	
31																	
32 33								11a						15c			
34																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.2b Carboxylic Acids														Copyright JAB chem	
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General			
27																	
28																	
29																	
30																	
31																	
32 33								11b									
34																	

Nat5 Traffic Lights			Past Paper Question Bank Unit 2.3 Energy From Fuels														Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027			
35		L9a	mc14	L6c	L3a	mc14		L9a		L11b(i)	L2c(ii)B	L1b(ii)	L11d(i)					
36	mc10	mc10											L5d(i)					
37		L8d			mc9	mc12	L16b			mc10	L2c(ii)A							
38	L9a L9c(i)				L9b(i)					L6b(i)		mc10	mc16 L5a					
39a							mc15	L9b(ii)		L6b(iv)		mc17						
39b																		
40	L9b(i)	L9b(i)	L9c	L8b	L9c	L14b	L9b	L9b(i)	L8b	L6b(iii)	L9a(i)	mc16	L9c(ii)B					

Nat5

Traffic Lights

Past Paper Question Bank

Unit 2.3 Energy From Fuels

Copyright

JAB

chem

Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015
35	mc4			L2a		L3a(ii)	L12a			L3a	L2a			mc1	L5a	
36				L2c		mc9						mc9				
37		mc12 L2a		L10a	L13b				mc8		mc10		mc11	mc11	mc12	
38	L11a									L6a						L4a
39a																
39b																
40															L8c	

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.3 Energy From Fuels														Copyright JAB chem	
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit			
35							16c										
36 37			8a							6b	20b		6b				
38				10a													
39a																	
39b																	
40																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 2.3 Energy From Fuels														Copyright JAB chem	
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General			
35						9b				9a				17c			
36 37	12a 12b		6a	4a 16c	6b	17a		9b(i)	19d			5d		11d			
38					9a				19c								
39a																	
39b																	
40																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.1a Metallic Bonding														Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027		
1	mc14	mc15		mc15	mc4	mc15	mc23			mc15			mc17				
2			L2b				L15b(ii)			mc16		L6a					

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.1a Metallic Bonding															Copyright JAB chem	
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015		
1		mc3			mc6				mc7	mc6	mc4							
2				mc7 L12a				L3a				mc7						

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.1a Metallic Bonding														Copyright JAB chem	
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit			
1																	
2																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.1a Metallic Bonding														Copyright JAB chem	
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General			
1																	
2	18c																

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.1b Reaction of Metals														Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027		
3a																	
3b																	
3c		mc16			L3c(i)					L8a L8b	L3c	mc19					
4	mc15		mc15			L8a(i) L8a(ii)	mc17	mc19	L5		mc18 L3b(ii)		L12				
5													L1a(iii)				

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.1b Reaction of Metals															Copyright JAB chem	
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015		
3a																		
3b																		
3c													mc25	L3a	L1a			
4			mc24	mc19			L10a				mc24	mc27		L3c L14c L14d	mc25			
5	L15a L15b			L9a L9b L9c				L15b(i) L15b(ii)							L11b	L14a(ii) L14a(iii)		

Nat5 Traffic Lights Past Paper Question Bank Unit 3.1b Reaction of Metals Copyright JAB chem																
Outcome	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013		
	Credit	Credit	Credit	Credit	Credit	Credit	Credit	Credit	Credit	Credit	Credit	Credit	Credit	Credit		
3a																
3b		11a								15b						
3c	7a												13b(i)			
4	7b	2b 11b 11c							2b	15a 15c						
5																

Nat5 Traffic Lights Past Paper Question Bank Unit 3.1b Reaction of Metals Copyright JAB chem																
Outcome	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013		
	General	General	General	General	General	General	General	General	General	General	General	General	General	General		
3a																
3b																
3c			2a	14a				13b	21c	4d			7c	12b		
4	14b		12a 12c 16c	14b	3b			13a	13a(i)			3b	4b	12a 12c(i)		
5																

Nat5 Traffic Lights			Past Paper Question Bank Unit 3.1c Redox													Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027		
6a 8a							L11c	L8d			L11c						
6b 8b			L11a		L10c(i)	L4b L8b	L7b(ii)	L10b(ii)	mc21								
7																	
9	mc19	mc20	mc17	L9b(i)	L10c(ii)	L10a(ii)	mc18	L10b(iii)	L9b(iv)	L13	mc20	mc20	mc19				

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.1c Redox															Copyright JAB chem
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015	
6a 8a		L13a			mc22	L4c(i)	L3b										
6b 8b	L13d		L3a	L13a		mc28	L12b(ii)		L14b(i)	mc28 L14b				L15b	L12b	L13b	
7	mc17		L13b(i)		L12a												
9		L14a		L3b				L12b	L14b(ii)		mc28				mc29		

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.1c Redox														Copyright JAB chem	
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit			
6a 8a							10b			17d	17c						
6b 8b			13b	16b(ii)	16c	10a(i)	14c	6c	13b	17a		21b	16b(ii) 19a	14b			
7																	
9																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.1c Redox													Copyright JAB chem	
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General		
6a 8a																
6b 8b																
7																
9																

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.1d Extraction of Metals Copyright JAB chem													
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027
10		L11a	L12a								L10a(ii)				
11a 12a	mc16	mc17	mc16			mc16									
11b 12b					mc15							L6b			
11c 12c			L12c							mc17	L10a(i)				
13 14				mc7		mc4			L9b(i) L9b(ii)	mc7					
15		L11b													

Nat5

Traffic Lights

Past Paper Question Bank

Unit 3.1d Extraction of Metals

Copyright

JAB

chem

Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015
10	L2b	L6a										L12a		mc29	L14c	
11a 12a		L6b	mc23				mc28		mc28			mc28			mc30	
11b 12b													mc29	mc30		
11c 12c	L2c				mc30						mc29					
13 14		L7a		L12c					L4a L4b							
15			mc8		mc7	L14c	mc27	mc28 mc29 L12a	L4c(i)	mc10		L12b(ii)	L12a	L1b	mc8	mc30

Nat5 Traffic Lights Past Paper Question Bank Unit 3.1d Extraction of Metals Copyright JAB chem																
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit		
10		11d							16d							
11a 12a		11b	15a					17e	16c		16b(ii)			16a		
11b 12b			15a						16b		16b(ii)			16a		
11c 12c			15a						16b		16b(ii)		21a	16a		
13 14							17a			10a	13a	18a				
15																

Nat5 Traffic Lights Past Paper Question Bank Unit 3.1d Extraction of Metals Copyright JAB chem																
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General		
10												14b				
11a 12a																
11b 12b													18d			
11c 12c																
13 14	18a	12a		4b		15a		17a(i)			15b	4b		14a		
15		12b 12c				15c		17a(ii)		12a 12b				14b 14c		

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.1e Electrochemical Cells														Copyright JABchem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027		
16					L10a		L11d				L3d(ii)	L6d(ii)					
17							L11b	L10a				L6c(ii)					
18	mc17				L10b(i) L10b(ii)	L10a(i)	L11a	L10b(i)		mc18	mc19	L6d(i)					
19 20 21	mc18	mc18 mc19	mc18	mc16			mc19	mc18		mc19	L3d(i)	L6c(i)	mc18				
22a 23a							L11c		mc22								
22b 23b										mc20							

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.1e Electrochemical Cells															Copyright JAB chem	
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015		
16		L14b							L14a			L14d			mc28 L13a	L11c		
17																		
18				L3c														
19 20 21	mc24	mc25	L6c	L3a	mc27 mc29	mc26	mc26	mc27 L13b	mc26 mc27	mc27	L14b(i)	L14a L14b L14c	mc23	mc26 mc27	L13b(i) L13b(ii)	L11b(i) L11b(ii) L16b		
22a 23a													mc28					
22b 23b																		

Nat5 Traffic Lights Past Paper Question Bank Unit 3.1e Electrochemical Cells Copyright JAB chem																
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit		
16	14a		13c(ii)	19a(ii)		19b		19c(ii)				21c		14c(ii)		
17				19a(i)												
18																
19 20 21	14c			19a(iii)												
22a 23a			13c(i)								17d			14c(i)		
22b 23b		15a 15b(i) 15b(ii) 15c		16a(i) 16a(ii)		19a	10a			17b		21a	19b			

Nat5 Traffic Lights Past Paper Question Bank Unit 3.1e Electrochemical Cells Copyright JAB chem																
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General		
16			L11a(iii)	17b	15a		19a				19d	21a(i)				
17		15a(i)	11a(i)													
18				17a		12a(i)	19b					21a(ii)	16a			
19 20 21		15b(ii)		17d	15c	12a(ii)	5a 5b	15b(ii)	3a 3b	15b(i)	19c		16b	18a(ii)		
22a 23a		15b(i)			15b			L15b(i)			19a			18a(i)		
22b 23b																

Nat5 Traffic Lights			Past Paper Question Bank Unit 3.2 Plastics														Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027			
24																		
25			L4b			L13b(ii)	L2a(i)		L6b(i)		L6a(ii)		L10a(ii)B					
26																		
27									L6a(i)			L10d(ii)A	L10a(ii)A					
28																		
29	mc20	mc21	L4a	mc17	L2a	L13b(i)	L2a(i)	mc20	L6b(ii)	mc21	L6a(iii)	L10d(ii)B	L3a(i)					
30					L2b		L2b											

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.2 Plastics															Copyright JAB chem
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015	
24																	
25								L7a						L10b(ii)			
26																	
27								L7c									
28																	
29	L7b	L3a	L7c	mc14	mc15	L6c	mc19	mc14 L7b	L6b	mc18	mc15	L10a(i)		L10b(i)	L5b	mc21	
30																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.2 Plastics														Copyright JAB chem	
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit			
24 25 26									3a		10b(ii)						
27	3a					5a	16b	4a						6a			
28 29 30	12a		10b(i)	14b	10a	11a	8a			18a	10b(i)	13a	20a	6d			

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.2 Plastics														Copyright JAB chem	
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General			
24 25 26	2a 20b(i)									11e		4c	14b				
27	20b(ii)					14b(i)		10c	17a	11d	17b(iii)		14a				
28 29 30		10a			11b							15a(ii)					

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.3 Fertilisers														Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027		
31 32						mc17	mc20		L3b(i)	L3e(i)	L8c(i)	mc22	L6				
33					L6a(ii)				L3b(ii)		L8c(ii)						
34								L3b		mc9		mc21 L2c L2d					
35																	
36		L12b(i)	L7a(ii)	L5a	L1c(ii)	mc18			mc24		mc21						
37	L11b	L12b(ii)															
38 39		L12c	L7b(i)				mc21 L10c			mc23	mc21	mc23 L2b	mc20				

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.3 Fertilisers															Copyright JAB chem	
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015		
31 32	L4b				L6c			mc23							mc24	mc28		
33																		
34		mc18	mc19	mc16	mc26 L6b						mc18							
35																		
36																		
37																		
38 39																		

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.3 Fertilisers														Copyright JAB chem	
Outcome	2000 <a>Credit	2001 <a>Credit	2002 <a>Credit	2003 <a>Credit	2004 <a>Credit	2005 <a>Credit	2006 <a>Credit	2007 <a>Credit	2008 <a>Credit	2009 <a>Credit	2010 <a>Credit	2011 <a>Credit	2012 <a>Credit	2013 <a>Credit			
31 32		13a(ii)	1a	2a										11a(ii)			
33																	
34					18b		L11c	1b									
35			18c														
36	17a	16a					3c		18c		7a			11b			
37	17b		18d(ii)										12c				
38 39						15b(i) 15b(ii)	1a			14a(i) 14a(ii) 14a(iii)	7b	19b(i) 19b(ii)					

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.3 Fertilisers														Copyright JAB chem	
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General			
31 32		14b	17c		4b	19c(i)	12b(i)	4a	7b	16a(ii)	16a 16b	6b	8b	1b			
33	5b		17b				12b(ii)										
34	21b	6a				19b				16b		7	7b	17a			
35																	
36			15b	3a		3d		2c		2a			11a				
37																	
38 39	8b	14a(i) 14a(ii)	1c						1c					17b			

Nat5 Traffic Lights			Past Paper Question Bank Unit 3.4 Nuclear Chemistry													Copyright JABchem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027		
40								L4a				L11	L7a				
41 44a 46a			L1a L5a	L2a L2b			mc22 L12a		mc23		mc22 L10b(ii)						
42 44b 46b	L12c	L14c			mc18	L5b			mc23	L1a(i) L1a(ii)		mc24	L7c(i)				
43							mc22		mc23								
45			L5c					L8e	L11a		L10b(i)		mc21				
47								L4b(i)			L10b(iii)A						
48								L4b(iii)		mc23			L7b(ii)				
49						L5a(i)			L11b(i) L11b(iii)				L7d				
50	L12a	L14a	L5b	L2c(i)	mc19	L5a(ii)	L12b	L4b(ii)	L11b(ii)	L1b(i)	L10b(iii)B		L7c(ii)				
51 52	L12b	L14b		L2c(ii)	mc17					mc24 L1b(ii)							

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.4 Nuclear Chemistry															Copyright JAB chem
Outcome	H 2000	H 2001	H 2002	H 2003	H 2004	H 2005	H 2006	H 2007	H 2008	H 2009	H 2010	H 2011	H 2012	H 2013	H 2014	H 2015	
40			mc29					mc39		L17a		L12a				mc40	
41 44a 46a				L3b			L14a(i)	mc40	mc39	L16a L16b					mc40		
42 44b 46b		L1b	L2b	mc39	L2a		mc40	mc40 L3a	L5a	mc39	mc40	L12b	L4a		mc40 L7b	L8a	
43	mc29							mc40		mc39					mc40		
45	L4a	L1a				mc39				mc40	L6a			L12a	L7a		
47																	
48		mc30 g32b	L2c	mc38		L6b(ii)					L6c		L4b(i)	mc39	L7c(ii)	L8c(ii)	
49			L2a	L3a					L5b(i)			L12c(i)			L7c(i)		
50	mc30	L1c			L2b(ii)	mc40	mc39	L3b(ii)	mc40	L16c	L6b		L4c	L12b		L8c(i)	
51 52							L14a(ii)		L5b(ii)			mc40				L8b	

Nat5 Traffic Lights Past Paper Question Bank Unit 3.5a Common Chemical Apparatus Copyright JAB chem																
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027	
53		L10b L13a(i)	L7c(ii)	mc4 L7a	L8b		L14b(i)		L3c(ii)	mc25	L6c(i)	L8b(i)	mc23 mc25			

Nat5 Traffic Lights Past Paper Question Bank Unit 3.5a Common Chemical Apparatus Copyright JAB chem																
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015
53	mc21 L12a L12b		L8a		L3a	L14a		L9a	L2a	mc26		L3b	L11a			L3a

Nat5 Traffic Lights Past Paper Question Bank Unit 3.5a Common Chemical Apparatus Copyright JAB chem																
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit		
53																

Nat5 Traffic Lights Past Paper Question Bank Unit 3.5a Common Chemical Apparatus Copyright JAB chem																
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General		
53																

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.5b General Practical Techniques														Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027		
54a											L8e						
54b										L7a(ii)							
54c		mc25						mc21					L1a(ii)				
54d																	
54e							mc25	L11c					L1a(iii)				
54f						mc19			mc19	L8d			mc22				
54g																	
54h					L10c(iii)			L10c			L3d(iii)						
54i							L7b(iii)										
54j																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.5b General Practical Techniques															Copyright JAB chem	
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015		
54a																		
54b																		
54c																		
54d																		
54e																		
54f	L5a			mc20	mc25		mc25			L2b(ii)				mc25				
54g						L4a												
54h											L14b(ii)							
54i																		
54j																		

Nat5 Traffic Lights Past Paper Question Bank Unit 3.5b General Practical Techniques Copyright JAB chem																
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit		
54a				15b					17b(ii)				20a(ii)	8b		
54b																
54c			14a													
54d																
54e																
54f																
54g																
54h																
54i	16a						16a									
54j																

Nat5 Traffic Lights Past Paper Question Bank Unit 3.5b General Practical Techniques Copyright JAB chem																
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General		
54a				7a			13b(i)				13b(iii)			19c		
54b																
54c																
54d						19a										
54e																
54f																
54g																
54h																
54i																
54j																

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.5c Analytical Methods														Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027		
55		L13b				mc20			L7b(i) L7b(ii)				L10b(i) L10b(ii)				
56																	
57	mc6	L6a	L3c		L6a(i)		L5b	mc22	L7c(i)	L11b(ii)	mc24						
58		mc22						mc23	L8a(i)	L11a(i)	L3c	L1b(i)	L1a(i)				
59			mc20				mc24	mc24									
60				mc19	mc20						mc23	L8a(ii)					

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.5c Analytical Methods Copyright JAB chem															
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015	
55					L16a												
56																	
57									L13c	mc22				L1a(i)		mc27	
58		L9a(i)				L3a(i)	L12b(i)	L4a(ii)	mc25			L3c		L9b(ii)			
59		L12a(ii)				mc23			L13a	mc25		L5a	L13a		mc26	mc29	
60		mc24	mc25					mc24			mc25						

Nat5 Traffic Lights Past Paper Question Bank Unit 3.5c Analytical Methods Copyright JAB chem																
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit		
55																
56																
57																
58								1a			12a		1b	1c		
59																
60		5b 18a								4b			7b	2a 13a		

Nat5 Traffic Lights Past Paper Question Bank Unit 3.5c Analytical Methods Copyright JAB chem																
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General		
55																
56																
57																
58			12d 14b(iii)	13b 18a		4c	9a	9b(ii)	11a 15b(i)	4b	11a(i)	19b		16b(ii)		
59																
60					14c(i) 14c(ii)		13b(i)		7a		13b(i) 13b(ii)	6a	20c	19b		

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.5d Reporting Experimental Work														Copyright JAB chem	
Outcome	Original Specimen Paper	New Specimen Paper	Nat5 2014	Nat5 2015	Nat5 2016	Nat5 2017	Nat5 2018	Nat5 2019	Nat5 2020	Nat5 2022	Nat5 2023	Nat5 2024	Nat5 2025	Nat5 2026	Nat5 2027		
61																	
62										L7b(ii)							
63	L3b(i)			L1b				L3c(ii)		L7c							
64								L3c(ii)									
65		L13a(ii)	L13a	L15a	L12c		L14b(ii)			L7b(i)		L8b(ii)	mc24 L9c(ii)A				
66	L4b(ii) L9b(ii)	L5b(ii) L9b(ii)	L7b(ii)	L8a L14c	L9a		mc15				L9a(ii)		L9c(i)				

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.5d Reporting Experimental Work															Copyright JAB chem
Outcome	Int2 2000	Int2 2001	Int2 2002	Int2 2003	Int2 2004	Int2 2005	Int2 2006	Int2 2007	Int2 2008	Int2 2009	Int2 2010	Int2 2011	Int2 2012	Int2 2013	Int2 2014	Int2 2015	
61																	
62																	
63																	
64																	
65												L11c(ii)				L15b(ii)	
66		L8 L11c															

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.5d Reporting Experimental Work														Copyright JAB chem	
Outcome	2000 Credit	2001 Credit	2002 Credit	2003 Credit	2004 Credit	2005 Credit	2006 Credit	2007 Credit	2008 Credit	2009 Credit	2010 Credit	2011 Credit	2012 Credit	2013 Credit			
61																	
62																	
63																	
64																	
65			19a(i)		21b(i)		18a				19a						
66																	

Nat5 Traffic Lights		Past Paper Question Bank Unit 3.5d Reporting Experimental Work														Copyright JAB chem	
Outcome	2000 General	2001 General	2002 General	2003 General	2004 General	2005 General	2006 General	2007 General	2008 General	2009 General	2010 General	2011 General	2012 General	2013 General			
61																	
62																	
63																	
64																	
65																	
66																	